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This document contains a collection of ideas and techniques for producing attractive technical drawings with John Hobby's METAPOST language. I'm assuming that you already know the basics of the language, that you have it installed as part of your up to date TeX ecosystem, and that you have established a reasonable workflow that let's you write a METAPOST program, compile it, and include the results in your TeX document. If not, you might like to start at the METAPOST page on CTAN, and read some of the excellent tutorials, including `mpintro.pdf`. If you have already done this, please read on.

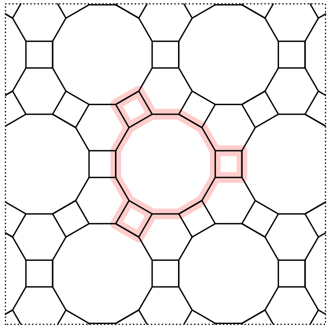
These notes are partly based on the examples I have developed as answers to questions about technical drawing on the TeX Stack Exchange site. In accordance with their terms and conditions, I've only included material here that I've written myself — if you want other people's code then visit the site; while most answers there focus on writing LaTeX documents, there are a great many questions about drawing, and some of the answers are very illuminating.

My approach here will be to explore plain METAPOST, with examples grouped into themes. One approach to using this document would be to read it end to end. Another would be to flick through until you see something that looks like it might be useful and then see how it's done.

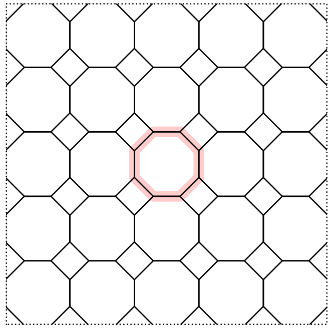
And when I say *plain* METAPOST I mean METAPOST with the default format (as defined in the file `plain.mp`) loaded and only a few simple external packages (like `colorbrewer`) occasionally. Nearly all of the examples here are supposed to be self contained, and any macros are defined locally so you can get to grips with what's going on. METAPOST is a very subtle language, and it's possible to do some very clever and completely inscrutable things with it; but here I have tried to be as clear as possible in my examples.

Drawing with Metapost

Toby Thurston — March 2017 – October 2024



$(4, 6, 12)$



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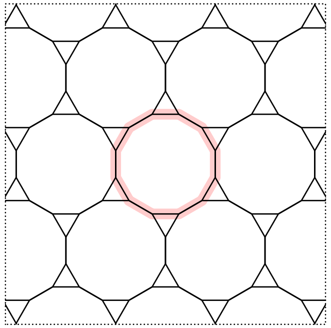

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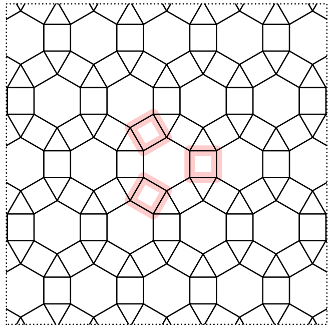

$(3, 4, 6, 4)$